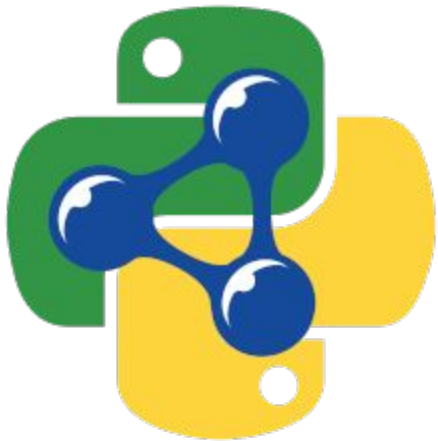




RDFLib

By Tom De Nies & Nicholas Car

Outline

- Quickstart - Jupyter demo
- Family of Tools
- Roles
- History
- Strengths & Weaknesses
- Future
- What do you want?
- Call for helpers



Quickstart

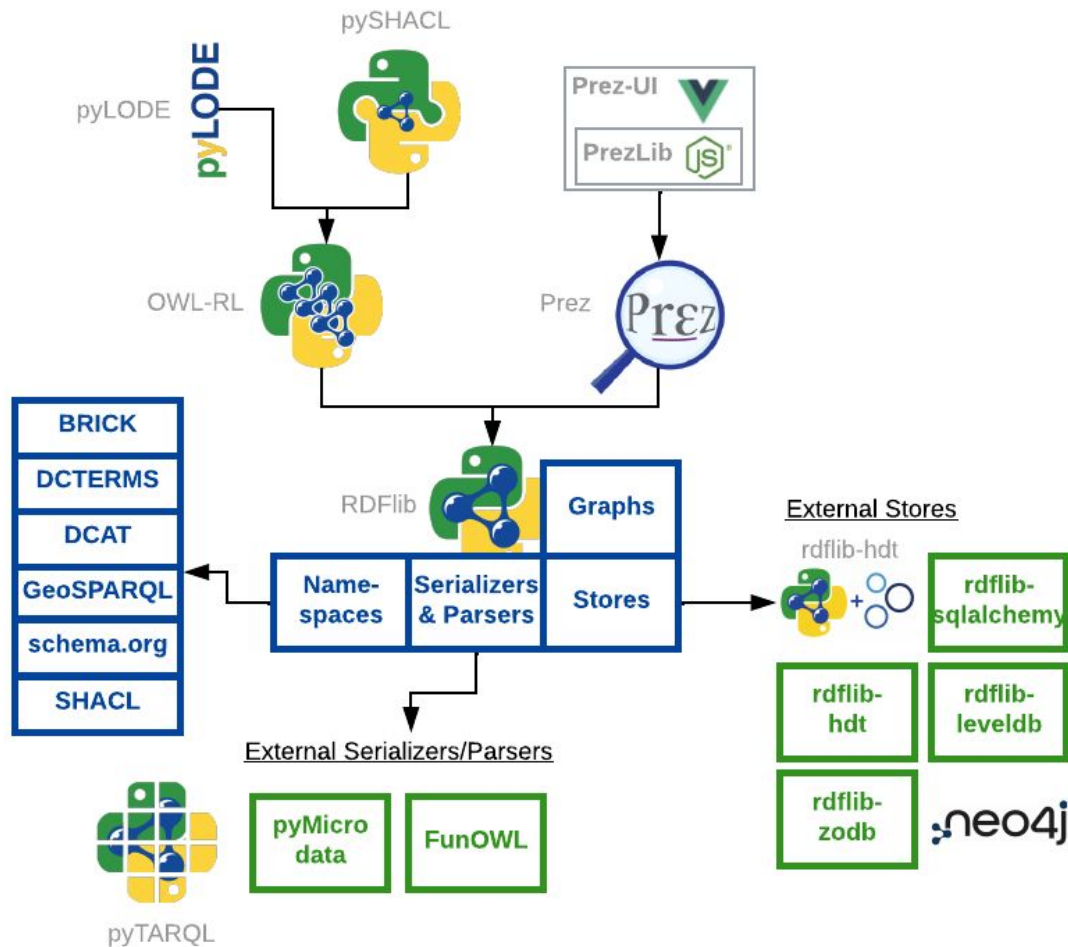
Some simple examples via Jupyter Notebooks online:

<https://github.com/RDFLib/notebook-demos>

Check out these examples: <https://github.com/RDFLib/rdflib/tree/main/examples>

Family of Tools

From <https://rdflib.dev>



Roles

1. Starting point

- For students
- For teaching, e.g. the previous Notebooks

2. Scripting

- Desktop use
 - working on 'my' RDF files
 - Supported by many extra packages & own scripts
 - AI now makes this even easier...
- Infracode
 - Use within GitHub Actions, Docker scripts, Functions apps

3. W3C Implementations

- For necessary implementations of W3C Semantic Web standards
 - OWL (OWL-RL), RDF, SPARQL, SHACL, PROF, SSN...

History

Content from <https://rdflib.readthedocs.io/en/latest/#history>

RDFLib is one of the oldest continuously maintained Python libraries for Semantic Web/RDF work. Its history goes back to 2002, only a few years after RDF itself emerged from the W3C. The original author was Daniel “eikeon” Krech but, of course, there have been many, many contributors to it since then.

Timeline

2002 — RDFLib begins

2004 — RDFLib 2.0

2005 — modern Graph abstraction emerges

2006 — SPARQL first supported

2010 — RDFLib 3.0
SPARQL separated into rdfextras

2013 — RDFLib 4.0; SPARQL 1.1 returns to core;
Dataset introduced

2017 — 4.2.2, followed by a long stable period

2020 — RDFLib 5.0; final Python-2-supporting major version

2021 — RDFLib 6.0; Python 3 only,
JSON-LD integrated

2023 — RDFLib 7.0

2026 — RDFLib 7.6,
with RDF4J/GraphDB client integrated

Strengths & Weaknesses

Strengths

- Easy
- Lots of extras
- Can use different Stores
- Widely used
- Well understood by AI

Weaknesses

- Some things are slow
 - Query parsing
 - RDF parsing
 - SHACL
- Not all extras are integrated*

** we are integrating more right now for v8!*

Future

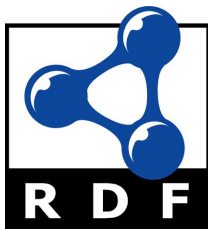
Near

- Current PRs
 - Lots of small things
 - Improved Turtle
 - Initial RDF 1.2
 - integrated reasoning
- Version 8
 - In 2026
 - More extras included
 - clients, CLI



RDF

- More reasoning
- RDF/SPARQL 1.2
- SHACL 1.2
- GeoSPARQL
- JSON-LD improved
 - pyLD integration
- New Formats
 - Jelly, Parquet
- New Stores
 - RocksDB, Oxygraph...



Python

- Slow growth of features
 - 20 yr code still works!
- New libraries
 - Lark parsers
- Rust / Performance
 - Critical points, optional



What do you want?

Tell us now or later!

We are seeing a rise in feedback now - Issues & PRs. Not sure why.

We want to integrate more extensions but we only know about a few of them.

Call for helpers

We do need more maintainers as the core is small:

- 5 very active
- 10 semi-active

Perhaps we do too much...

What can you help with *as a maintainer*?

1. Helping PRs pass tests
2. Rationalising duplicate PRs and closing issues
3. Encourage others to test RDF stuff in or on top of RDFLib - and tell the community!